6-2 Binary and Amount of Information

Point!

1 Bits and Bytes

(1) ([1]**Bit**) : The smallest unit of information, having only two possible states: "0" and "1". One bit is capable of representing two states; for example, "a switch is on or off," "voltage is high or low," "or the orientation of a magnet is north or south."

- Actual information is represented by the arrangement of bits. In general, n bits can represent ([2]2^n) different pieces of information.

<Example> 1 bit : 2 possibilities: 0 or 1
2 bits : 4 possibilities: 00, 01, 10, 11 — $2^2 = 4$ possibilities
3 bits : 8 possibilities: 000, 001, 010, 011, 100, 101, 110, 111 — $2^3 = 8$ possibilities

(2) ([3]**Byte**) : A unit consisting of 8 bits. The notation is ([4]**B**).
1 B = ([5]**8 bits**), which can represent ([6]**256**) possibilities. — $2^8 = 256$ possibilities

(3) Unit of information : The basic unit is 1 B. The unit changes every 2^{10} = ([7]**1,024**) times.
[1] 1 ([8]**KB**) = 1,024 B
[2] 1 ([9]**MB**) = 1,024 KB
[3] 1 ([10]**GB**) = 1,024 MB
[4] 1 ([11]**TB**) = 1,024 GB

2 Decimal and Binary

(1) ([12]**Decimal**) : A method of representing numbers using ten digits from "0" to "9". A number expressed in decimal notation is denoted as ([13]**decimal**).

(2) ([14]**Binary**) : A method of representing numbers using two types of digits, "0" and "1". A number represented in binary is sometimes written with a subscript (2) placed at the bottom right of the number, such as ([15]**binary number**).

(3) Conversion between Decimal and Binary
[1] Conversion from binary to decimal
Starting from the rightmost digit, multiply each digit by 1, 2^1, 2^2, 2^3, ... in order, and then sum the results.
<Example> Convert the binary number $1011_{(2)}$ into decimal form.
$1011_{(2)} = (1 \times 2^0 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3) = 11$

[2] Conversion from decimal to binary
Take a decimal number and keep dividing it by 2 until the quotient becomes 1, then arrange the resulting quotients and remainders in reverse order from the left.
<Example> Represent the number "6" in decimal as a number in binary.

$$\begin{array}{l} 2\overline{)6} \\ 2\overline{)3} \cdots 0 \\ \quad 1 \cdots 1 \end{array}$$

Therefore, 6 = ([17]**$110_{(2)}$**)

Warm Up

Answer the following questions.

(1) [1] Complete the following sentences by filling in the blanks **A** to **E** with the appropriate terms and numbers.

> The smallest unit of information is called a bit, which corresponds to (**A**) digit in binary notation. This makes it possible to represent (**B**) different types of information. Additionally, a group of (**C**) bits is called one (**D**), and is represented as 1 B. For example, 24 bits is (**E**) B.

[2] How many bits of information are required to represent all possible outcomes when throwing a set of dice (one large and one small)?

[3] How many B is 1 MB? Answer in the form of a power of 2.

[4] How many times greater is 4 bits of information compared to 2 bits of information?

(2) [1] Express the following binary numbers in decimal form.

(A) $11010_{(2)}$ (B) $101011_{(2)}$

[2] Express the following decimal number in binary.

(A) 39 (B) 120

Explanation

(1) [1] **A:** A single digit in binary can represent two different values. 1 **B:** 2 **C:** 8
D: Byte **E:** Since 8 bits = 1 B, 24 bits ÷ 8 = 3 B

[2] When you roll two dice (one large and one small), there are 6 × 6 = 36 possible combinations of outcomes.
With 5 bits, you can represent $2^5 = 32$ combinations,
and with 6 bits, you can represent $2^6 = 64$ combinations,
so the needed information is 6 bits

With n bits, it is possible to represent 2^n different pieces of information.

[3] 1 MB = 1,024 KB, and 1 KB = 1,024 B,
therefore, 1 MB = 1,024 KB = 1,024 × 1,024 B = $2^{10} \times 2^{10}$ B = 2^{20} B

[4] The amount of information that can be represented with 2 bits is $2^2 = 4$ possibilities, and the amount of information that can be represented with 4 bits is $2^4 = 16$ possibilities. Therefore, 16 ÷ 4 = 4 times

(2) [1] (A) $11010_{(2)} = 0 \times 1 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 1 \times 2^4$
$= 0 + 2 + 0 + 8 + 16 = 26$

(B) $101011_{(2)} = 1 \times 1 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 0 \times 2^4 + 1 \times 2^5$
$= 1 + 2 + 0 + 8 + 0 + 32 = 43$

[2] (A)

2) 39
2) 19 ⋯1
2) 9 ⋯1
2) 4 ⋯1
2) 2 ⋯0
1 ⋯0

Therefore, $100111_{(2)}$

Add (2) to the bottom right

(B)

2) 120
2) 60 ⋯0
2) 30 ⋯0
2) 15 ⋯0
2) 7 ⋯1
2) 3 ⋯1
1 ⋯1

Therefore, $1111000_{(2)}$

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [3] with the appropriate terms and numbers.

> The numbers we use in our daily lives are expressed in (　[1]　), using digits from 0 to 9. There is also (　[2]　) using the digits 0 and 1, and the information handled by computers primarily utilizes ([2]). However, because this unit is too small and difficult to understand, one byte, which consists of (　[3]　) bits, is often used to represent information.

(2) How many bits are in 5 bytes?

(3) How many different pieces of information can be represented by 1 byte?

(4) How many times is the amount of information in 5 bits compared to the amount of information in 3 bits?

(5) How many MB are in 1 GB? Also, how many B is it? Answer in the form of a power of 2.

(6) How many bits of information are needed to represent the four seasons of spring, summer, autumn, and winter?

(7) How many bits of information are needed to represent a deck of 52 playing cards, excluding the jokers?

(8) How many bits of information are needed to record all possible outcomes when tossing a coin three times?

2 Answer the following questions.

(1) Express the following binary numbers in decimal form.

[1] $110_{(2)}$　　[2] $10100_{(2)}$　　[3] $111001_{(2)}$

(2) Express the following decimal number in binary.

[1] 65　　[2] 106　　[3] 143

♪ **3** How many 4.7 GB DVDs worth of data can be stored on a 1 TB hard disk? Round your answer to the nearest whole number.

Exercise

1 Answer the following questions.

(1) Choose the number or equation that best fit into the blanks [1] to [6] from the options **A** to **J**.

> The amount of information that can be represented by 1 bit is ([1]) possibilities. Additionally, since 1 byte consists of ([2]) bits, the amount of information that can be represented by 1 byte is ([3]) possibilities. For example, 32 bits equal ([4]) bytes. Additionally, 1 KB is expressed as ([5]) B. To calculate how many bits are in 24 KB, you need to calculate the expression ([6]).

A 1	**B** 2	**C** 4	**D** 8
E 256	**F** 1,024	**G** 24 × 1,024	**H** 1,024 ÷ 24
I 24 × 1,024 × 8	**J** 24 × 1,024 ÷ 8		

(2) Choose one option that represents the correct order of the amount of information from the options **A** to **D** below.

A 1 KB < 1 MB < 1 GB < 1 TB
B 1 KB < 1 MB < 1 TB < 1 GB
C 1 TB < 1 KB < 1 MB < 1 GB
D 1 KB < 1 TB < 1 MB < 1 GB

(3) How many different combinations of designs can be created using one 100 yen coin, one 50 yen coin, and one 10 yen coin? Also, how many bits of information does this represent?

(4) How many bits of information are needed to represent 16 directions?

(5) What is the minimum number of bits needed to represent the 47 prefectures?

2 Answer the following questions.

(1) Express the following binary numbers in decimal form.

[1] $101_{(2)}$ [2] $10101_{(2)}$ [3] $101101_{(2)}$

(2) Express the following decimal number in binary.

[1] 13 [2] 128 [3] 138

3 How many photos, each with a size of 2 MB, can be stored on a 32 GB USB flash drive?